

A01 - Full Validation

Host-side logical NVMe validation framework with deterministic fault injection, queue lifecycle tracking, recovery validation, and firmware-log correlation.

STOPPED

9	5	4	0
TOTAL	EXECUTED	PASSED	FAILED
0	1	4	1.331 s
ERROR	ABORTED	NOT RUN	DURATION

Suite ID	A01
Suite Name	Full Validation
Generated	2026-09-22 01:23:24
Storage Baseline Reset	true
Stop Reason	User requested stop
Execution State	5 executed / 4 not run

01

Test Summary

ID	Test	Result	Duration
T01	Device Baseline Check	PASS	0.105 s
T02	Normal Write / Readback	PASS	0.104 s
T03	Invalid Read Range	PASS	0.054 s
T04	Unsupported Command	PASS	0.055 s
T05	Read Timeout + Recovery	ABORTED	1.013 s
T06	Data Integrity Failure	NOT_RUN	0.000 s
T07	Firmware Log Parsing	NOT_RUN	0.000 s
T08	Read Error Correlation	NOT_RUN	0.000 s
T09	Write Failure + Recovery	NOT_RUN	0.000 s

02

Detailed Validation Results

T01

Device Baseline Check

PASS

Expected	Device baseline matches configuration
Actual	Baseline validation passed
Validation Checks	10 / 10 PASS
Duration	0.105 s

Key Validation	Expected	Actual	Status
Firmware Revision	1.0.3	1.0.3	PASS
Capacity Blocks	4096	4096	PASS
Logical Block Size	512	512	PASS
Temperature	<= 70	35	PASS
Critical Warning	0	0	PASS
Media Errors	0	0	PASS

T02 Normal Write / Readback**PASS**

Expected	Write and read back T02_WRITE_READBACK at LBA 600
Actual	Write / Readback validation passed
Validation Checks	6 / 6 PASS
Duration	0.104 s

Key Validation	Expected	Actual	Status
Completion Status	SUCCESS	SUCCESS	PASS
Readback LBA	600	600	PASS
Readback Length	1	1	PASS
Readback Data	T02_WRITE_READBACK	T02_WRITE_READBACK	PASS
Unwritten Flag	false	false	PASS

T03 Invalid Read Range**PASS**

Expected	READ LBA 4095, Length 2 returns INVALID_RANGE
Actual	Invalid range correctly rejected
Validation Checks	4 / 4 PASS
Duration	0.054 s

Key Validation	Expected	Actual	Status
Completion Status	INVALID_RANGE	INVALID_RANGE	PASS
Completion Error	INVALID_RANGE	INVALID_RANGE	PASS
Completion Data	None	None	PASS
Timed Out	false	false	PASS

T04 **Unsupported Command****PASS**

Expected	FORMAT returns UNSUPPORTED_COMMAND
Actual	Unsupported command correctly rejected
Validation Checks	5 / 5 PASS
Duration	0.055 s

Key Validation	Expected	Actual	Status
Completion Status	UNSUPPORTED_COMMAND	UNSUPPORTED_COMMAND	PASS
Completion Error	UNSUPPORTED_COMMAND	UNSUPPORTED_COMMAND	PASS
Completion Data	None	None	PASS
Timed Out	false	false	PASS
Submitted Opcode	FORMAT	FORMAT	PASS

T05 **Read Timeout + Recovery****ABORTED**

Expected	Initial READ times out, Controller Reset clears queue state, and retry succeeds with a new CID
Actual	Test aborted at INITIAL_READ: User requested stop
Validation Checks	No completed validation checks
Duration	1.013 s
Abort Stage	INITIAL_READ
Stop Reason	User requested stop

T06 **Data Integrity Failure****NOT_RUN**

Expected	Scheduled for execution
Actual	Not run because suite stop was requested
Validation Checks	Not executed
Duration	0.000 s

T07 **Firmware Log Parsing**

NOT_RUN

Expected	Scheduled for execution
Actual	Not run because suite stop was requested
Validation Checks	Not executed
Duration	0.000 s

T08 **Read Error Correlation**

NOT_RUN

Expected	Scheduled for execution
Actual	Not run because suite stop was requested
Validation Checks	Not executed
Duration	0.000 s

T09 **Write Failure + Recovery**

NOT_RUN

Expected	Scheduled for execution
Actual	Not run because suite stop was requested
Validation Checks	Not executed
Duration	0.000 s

03

Failure, Recovery, and Stop Evidence

STOP v1 - Suite Execution Control	
Suite Result	STOPPED
Stop Reason	User requested stop
Executed	5 / 9
Aborted Tests	T05
Not Run	4
Interpretation	The active unit was cooperatively cancelled; remaining scheduled units were not started.

T05 - READ Abort + Cancellation Recovery	
Abort Stage	INITIAL_READ
Stop Reason	User requested stop
Command	CID 1 / READ / LBA 100
Host Result	ABORTED
Timed Out	false
Process Termination	terminate
Outstanding After Abort	[]
Queue After Abort	SQ 0/0, CQ 0/0
Next CID	2

Technical Notes and Scope

Queue Model	The project models Submission Queue (SQ), Completion Queue (CQ), CID allocation, queue head/tail progression, and logical doorbell events on the Host side.
Mock Controller	The controller executes in a separate Python subprocess and receives Host-assigned commands and CIDs.
Fault Injection	Timeout, NAND_READ_FAIL, NAND_PROGRAM_FAIL, and data miscompare faults are deterministic and designed for validation workflow demonstration.
Firmware Log	Runtime firmware-style log entries are generated by the Mock Controller process and can be correlated with command results by CID, LBA, status, and error.
STOP v1	Cancellation is cooperative. Active fake_nvme subprocesses can be terminated and recovered through simulated controller reset; arbitrary Python deadlocks or non-cooperative infinite loops are outside the v1 stop guarantee.
Hardware Scope	This project is a Host-side logical NVMe validation simulation. It does not emulate or claim real PCIe MMIO, DMA, PRP/SGL, MSI/MSI-X, FTL, NAND media behavior, or physical SSD firmware validation.

Purpose: demonstrate validation design, automation structure, failure handling, recovery logic, queue-state reasoning, and evidence correlation in a deterministic mock environment.